ODAS Concurrent Stress Test

Four versions launched in sequence, then measured under simultaneous load

Purpose. Measure per-stage processing time and CPU load when four ODAS snapshots run together. The snapshots have different features, so this is a concurrent-load test, not a comparison of identical code paths.

Test sequence

Prebuilt Release binaries ran on ARM64 using the live `multi_capture` input. Start the first version, wait five seconds between each later start, then keep all four running together for 15 seconds. The first five seconds at full overlap are warm-up; collect measurements over the final 10 seconds and stop all four together.

Elapsed	Action	Active
0 s	Start bcb8454	1
5 s	Start c603141	2
10 s	Start 18a8a73	3
15 s	Start 93a1b81; all-four overlap begins	4
20 s	Final 10-second measurement begins	4
30 s	Stop all four	0

Final 10-second results

Version	Frames	SSL ms/frame	SST ms/frame	SSL + SST	CPU (% of one core)
bcb8454	1,250	1.002	0.010	1.012 ms	19.4%
c603141	1,262	1.700	0.013	1.713 ms	36.3%
18a8a73	1,235	1.659	0.015	1.674 ms	41.8%
93a1b81	1,273	1.565	0.013	1.578 ms	42.6%

Concurrent versions	Host CPU	ODAS CPU sum
1	23.6%	29.4%
2	19.5%	64.6%
3	31.7%	107.8%
4	41.2%	143.5%
4, final window	40.7%	140.0%

How to read CPU: process CPU uses 100% per logical core. Thus, 140% combined is about 1.4 cores, not 140% of the machine. The host reports four logical CPUs; host CPU averaged 40.7% during the final window.

Summary

SSL dominates processing in every version; SST is only 0.010–0.015 ms/frame. Combined SSL + SST ranged from 1.012 ms/frame (bcb8454) to 1.713 ms/frame (c603141); 93a1b81 measured 1.578 ms/frame. The four processes used about 1.4 cores together.

The largest current-version SSL stage was Freq at Peak (0.541 ms/frame). Activity + Classify was about 0.001 ms/frame, which does not establish standalone YAMNet inference time. These are module timings from one 10-second window, not end-to-end latency, worst-case timing, or a statistical benchmark.

Full results: ODAS_Concurrent_Stress_Report.md